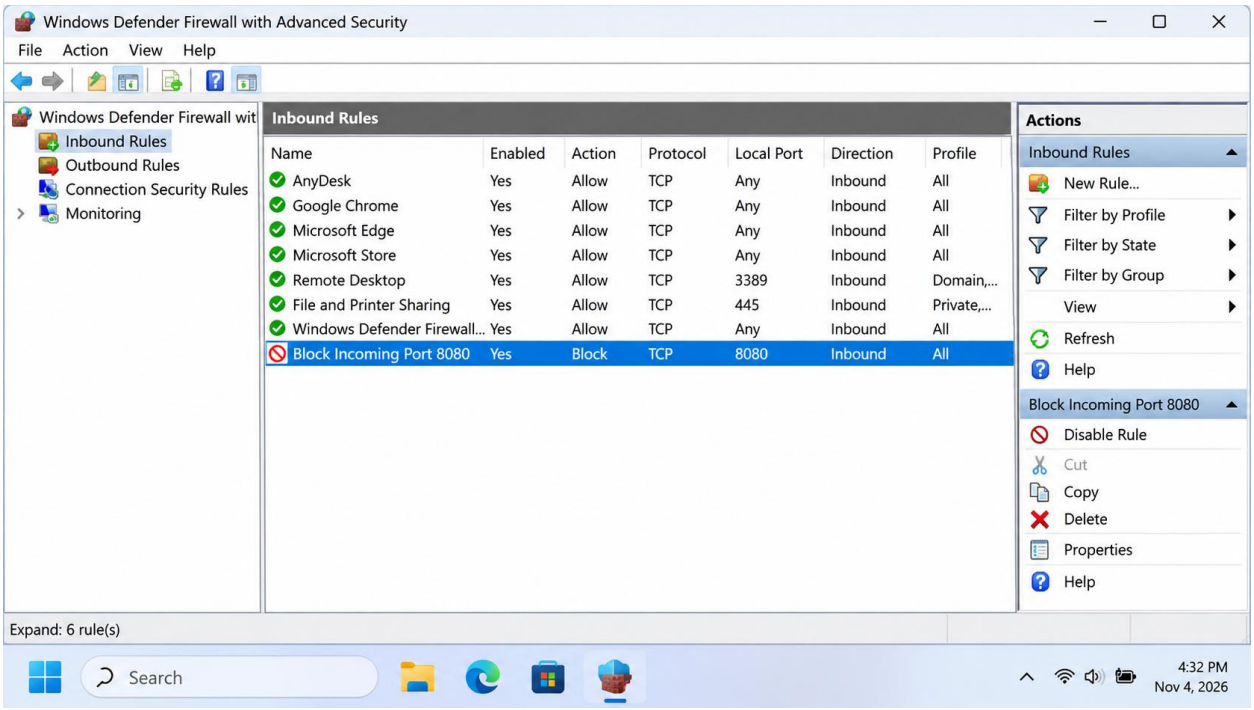
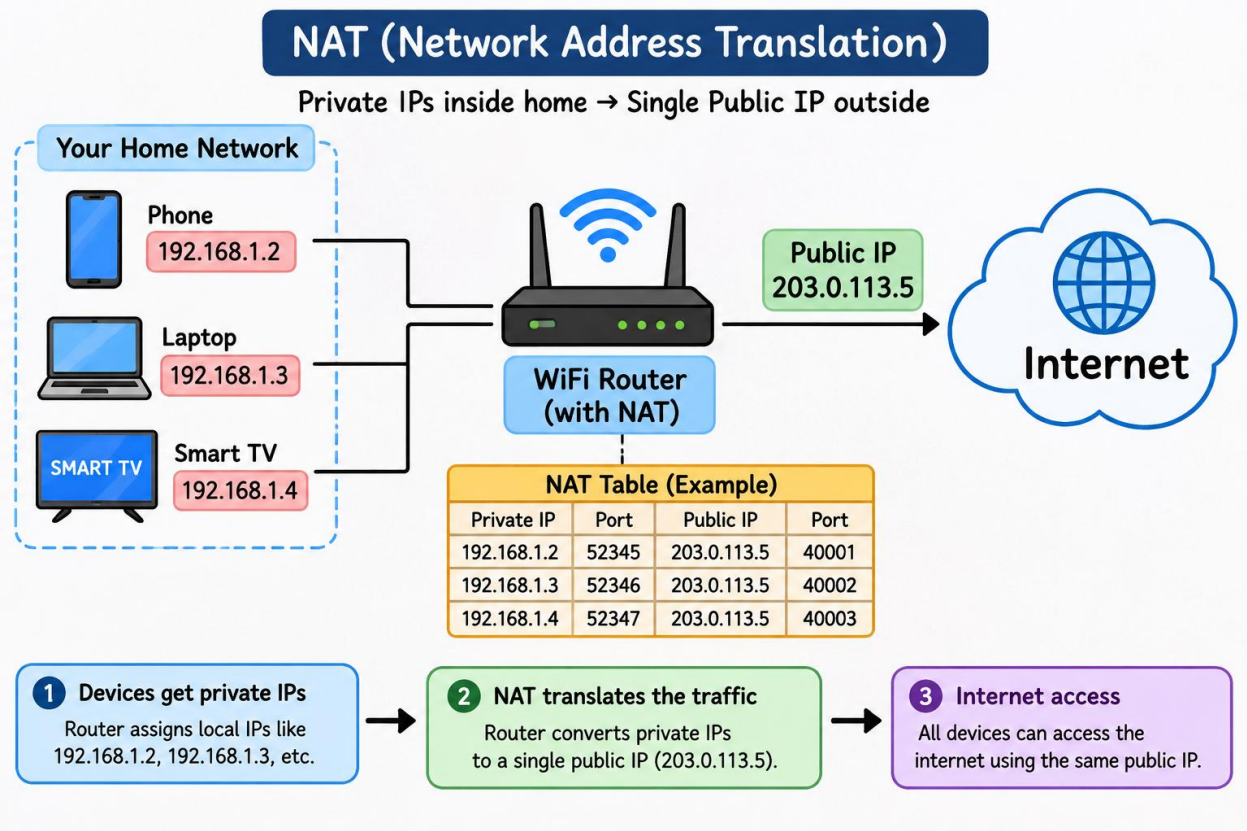
1)Set up Windows Defender Firewall on your Windows laptop or PC and create a new inbound rule to block all incoming traffic on port 8080. Take a screenshot of the rule you created.



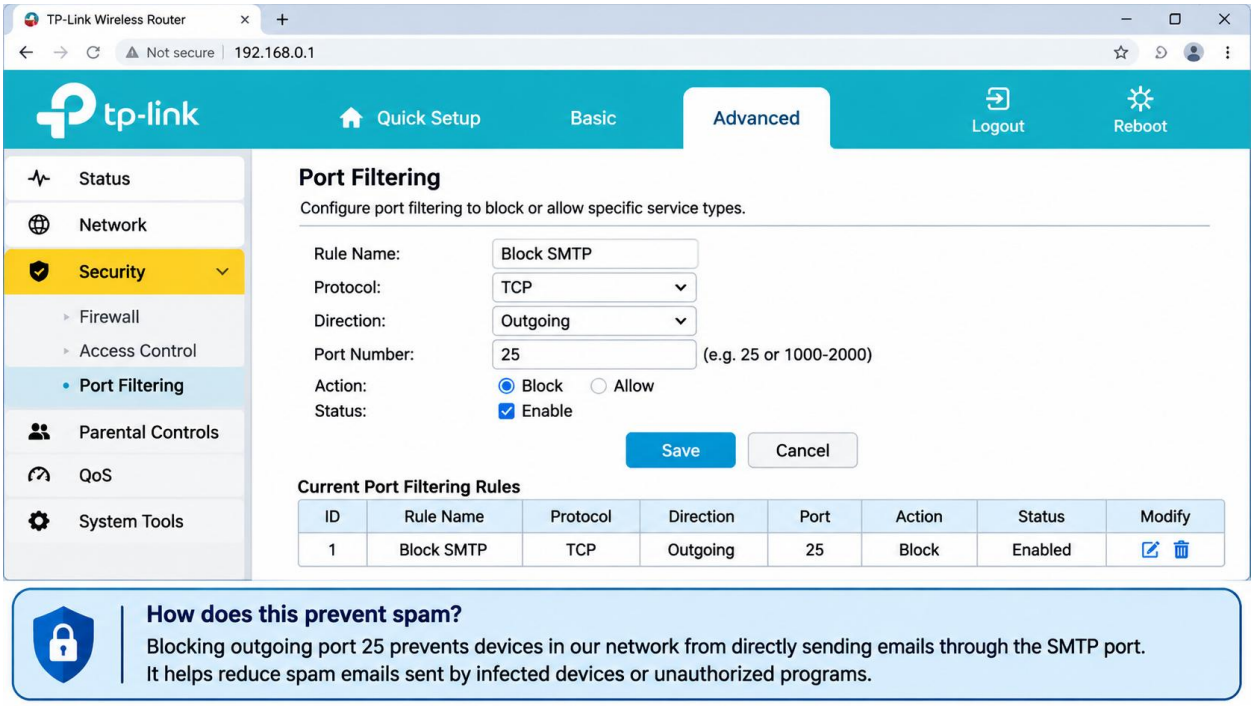
2)Simulate NAT (Network Address Translation) by drawing a simple diagram showing how your home WiFi router assigns private IP addresses to your phone, laptop, and smart TV, and how it translates them to a single public IP for internet access. Use any drawing tool or even hand-draw and click a photo.

Ans.



3)Configure port filtering using your WiFi router's admin panel (or use a router simulator online if you don't have access). Block outgoing traffic on port 25 (SMTP) and explain in 2 sentences how this could help prevent spam from your network.

Ans.



4)Given a sample packet filtering rule set (provided by your trainer), identify which of these three sample packets would be allowed or blocked: 1) HTTP request from 192.168.1.5 to 172.217.166.78:80, 2) SSH request from 10.0.0.10 to 34.201.23.1:22, 3) DNS query from 192.168.1.7 to 8.8.8.8:53.Focus on source/destination IPs and ports in the rule set.

Ans. To decide whether the packets are Allowed or Blocked, we need to check their source IP, destination IP, and destination port against the firewall rules.

Packet Filtering Analysis

No.	Source IP	Destination IP	Port	Service
1	192.168.1.5	172.217.166.78	80	HTTP
2	10.0.0.10	34.201.23.1	22	SSH
3	192.168.1.7	8.8.8.8	53	DNS

Explanation

HTTP (Port 80): Used for accessing websites.

SSH (Port 22): Used for secure remote access to computers.

DNS (Port 53): Used to convert website names into IP addresses.

5)Research and list 3 ways Instagram or Zomato might use firewalls and packet filtering to protect their servers from attacks or misuse.

Ans. 3 Ways Instagram or Zomato Use Firewalls and Packet Filtering to Protect Their Servers

1. Blocking Unwanted Traffic: Instagram and Zomato can use firewalls to block suspicious traffic and unauthorized connections from unknown sources. This helps protect their servers from hackers.

2. Allowing Only Safe Connections: They can use packet filtering to allow trusted network traffic and block unwanted IP addresses or ports. This helps protect important data and services.

3. Preventing Network Attacks: Firewalls can help detect and block harmful traffic, such as repeated suspicious requests or traffic from malicious sources. This helps keep their apps and websites safe and available.